AN INVERTIBLE MODULE ON A FINITE COVER IS TRIVIAL OVER THE PREIMAGE OF A NEIGHBOURHOOD OF ANY POINT OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-finite-trivialize-invertible-upstairs is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 3616-3622 and proof span 3624-3652 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof contains the author's sentence offering a cited alternative ('This follows from Properties, Lemma ... but we also give a direct argument.'); both halves are preserved and the direct argument is the one counted.

Editorial note. 2 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A finite morphism is closed and has finite fibres, and the fibre of a finite morphism over a point of an affine base is finite. Used at proof lines 3625-3626 and at 3649-3651. Gives both the finiteness of the fibre used to select the auxiliary set of closed points and the closedness used to push the open set down.
- Surjectivity of a map of quasi-coherent modules over an affine scheme after taking global sections, and the identification of global sections with the sum of the fibres at a finite set meeting every specialization class. Used at proof lines 3633-3643 with the cited schemes chapter lemma on the equivalence of quasi-coherent modules. Produces a global section of the invertible module that does not vanish at any point of the fibre.
- Existence of a section of an invertible module on an affine scheme not vanishing at a prescribed finite set of points, cited from the properties chapter as the alternative to the direct argument given here. Cited at proof lines 3628-3630. The author offers it as the shortcut and then gives the direct argument that is counted here.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma properties-lemma-quasi-affine-invertible-nonvanishing-section: sources/stacks-divisors/pinned-original-source.tex, source lines 3241–3241. Same-chapter reference used in the retained author text at source line(s) 3241, 3630. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma schemes-lemma-equivalence-quasi-coherent: sources/stacks-divisors/pinned-original-source.tex, source lines 3645–3645. Same-chapter reference used in the retained author text at source line(s) 3645. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $\pi : X \to Y$ be a finite morphism of schemes. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Let $y \in Y$. There exists an open neighbourhood $V \subset Y$ of y such that $\mathcal{L}|_{\pi^{-1}(V)}$ is trivial.*

Proof. Clearly we may assume Y and hence X affine. Since π is finite the fibre $\pi^{-1}(\{y\})$ over y is finite. Since X is affine, we can pick $s \in \Gamma(X, \mathcal{L})$ not vanishing in any point of $\pi^{-1}(\{y\})$. This follows from Properties, Lemma properties-lemma-quasi-affine-invertible-nonvanishing-section but we also give a direct argument. Namely, we can pick a finite set $E \subset X$ of closed points such that every $x \in \pi^{-1}(\{y\})$ specializes to some point of E. For $x \in E$ denote $i_x : x \to X$ the closed immersion. Then $\mathcal{L} \to \bigoplus_{x \in E} i_{x,*} i_x^* \mathcal{L}$ is a surjective map of quasi-coherent $\mathcal{O}_X$-modules, and hence the map

$$\Gamma(X, \mathcal{L}) \to \bigoplus\nolimits_{x \in E} \mathcal{L}_x / \mathfrak{m}_x \mathcal{L}_x$$

is surjective (as taking global sections is an exact functor on the category of quasi-coherent $\mathcal{O}_X$-modules, see Schemes, Lemma schemes-lemma-equivalence-quasi-coherent). Thus we can find an $s \in \Gamma(X, \mathcal{L})$ not vanishing at any point specializing to a point of E. Then $X_s \subset X$ is an open neighbourhood of $\pi^{-1}(\{y\})$. Since π is finite, hence closed, we conclude that there is an open neighbourhood $V \subset Y$ of y whose inverse image is contained in X_s as desired. □